Are Apple's Quarterly Returns Forecastable?

A Time-Series Investigation, 1981–2026

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The research question

Question

Do Apple's quarterly stock returns contain stable temporal dependence and seasonality that are strong enough to produce out-of-sample forecasts that outperform simple naive benchmarks on recent quarters?

What a "yes" would look like: an ARIMA or regularized regression beats both the historical-mean and persistence baselines on MAE, RMSE, and a Diebold–Mariano test, with white-noise residuals.
What a "no" would look like: models tie or trail the naive benchmarks; gains are not stable across nearby test windows; residuals still carry structure.

Why Apple, why quarterly?

- **Single firm, not an index.** Avoids cross-sectional aggregation; isolates one company's earnings cycle.
- **Quarterly cadence.** Aligns with Apple's reporting calendar and with product-launch patterns (iPhone in Q4, education in Q3).
- **Long sample.** 1981Q1–2026Q2 spans the entire public history of Apple as a listed firm.
- **Tractable size.** 182 observations is enough for ARIMA but small enough to expose the limits of complex models.

Why a time-series problem?

- Returns are *ordered* in time; OLS independence assumption does not hold by default.
- Seasonal effects (fiscal-quarter dummies) and lagged returns are natural candidate features.
- Forecasting is a *strictly causal* task: at quarter t, only information available before t may be used.
- Ignoring temporal dependence gives biased standard errors and optimistic out-of-sample claims.

Talk roadmap

1. **Data** — the Apple quarterly panel and the target variable.
2. **EDA & diagnostics** — stationarity, autocorrelation, seasonality, conditional heteroskedasticity.
3. **Models** — naive benchmarks, ARIMA / SARIMA, lagged regressions with regularization.
4. **Evaluation** — holdout, rolling-origin backtest, Diebold–Mariano test.
5. **Conclusion** — answer, limitations, future work.

What success and failure each look like

Positive (yes)

- Lower RMSE *and* MAE than every naive benchmark
- Diebold–Mariano $p < 0.05$
- White-noise residuals (Ljung–Box not rejected)
- Stable across rolling windows

Null (no)

- Errors comparable to naive mean
- DM tests not significant
- Residual structure remaining
- Gains not stable

Pre-registering both outcomes prevents post-hoc story-fitting.

Dataset at a glance

Source. Kaggle "Apple Financial Dataset (1980–2026)".

- `aapl_quarterly_master.csv` — primary table, **182 quarters**, 1981Q1 – 2026Q2.
- `aapl_quarterly_summary.csv` — product revenue shares (only complete from 2009Q1; used for sensitivity).
- Daily-level files used only for context.
- No missing values in the target.
- Quarter index built from `(fiscal_year, fiscal_quarter)` pairs; no duplicates, strictly monotonic.

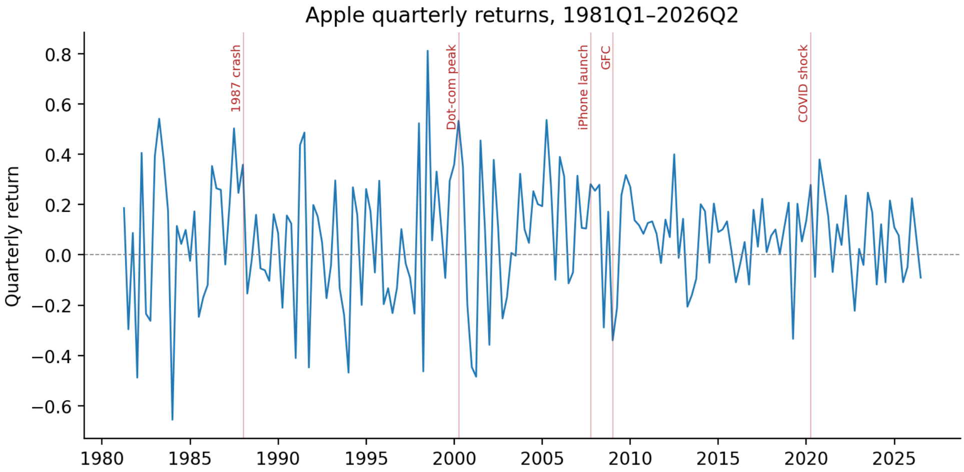
The target variable: quarter_return

Definition. `quarter_return` is Apple's compounded return over each fiscal quarter, computed from intra-quarter daily returns.

Statistic	Value
n	182
Mean	0.067
Std. dev.	0.236
Min	-0.656
Max	$+0.812$

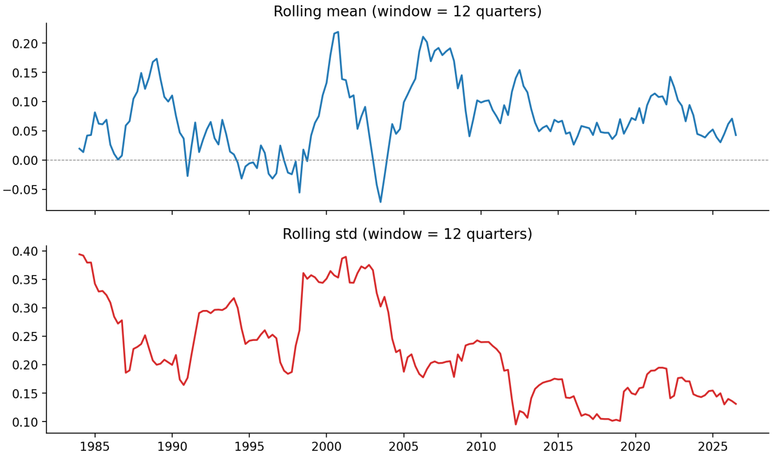
Robustness target: `price_return_pct` (open-to-close, $\rho = 0.99$).

Forty-five years of quarterly returns



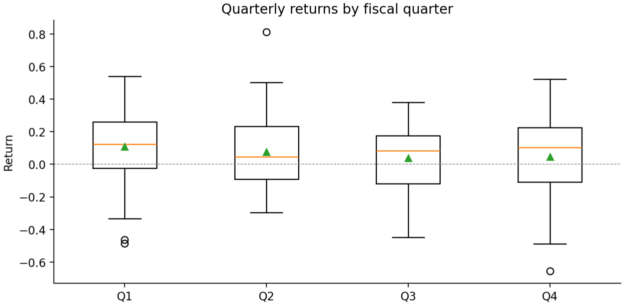
Vertical lines: 1987 crash, dot-com peak, GFC, COVID shock, iPhone launch.

Rolling mean and rolling volatility



Twelve-quarter rolling mean is roughly flat near zero; rolling standard deviation is decisively *not* flat.

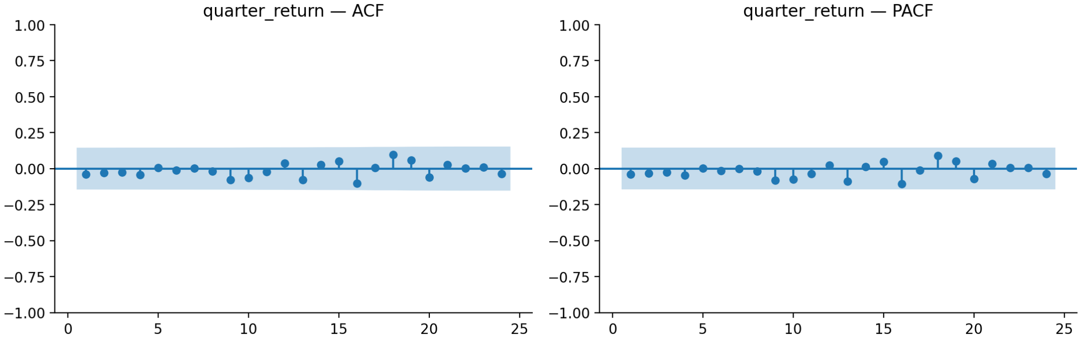
Seasonal pattern by fiscal quarter



Quarter	Mean	Std
Q1	0.108	0.236
Q2	0.076	0.237
Q3	0.038	0.197
Q4	0.046	0.270

Means differ across quarters but within-quarter variability is roughly 5$\times$ the between-quarter spread. Seasonality is real but weak.

ACF and PACF



No spike pierces the 95% confidence band at any lag up to 24. Sample ACF at lag 1 is -0.04.

Stationarity tests agree

Test	Statistic	p-value	Conclusion
ADF (H_0: unit root)	-13.90	$< 10^{-25}$	stationary
KPSS (H_0: stationary)	0.09	0.10	stationary

Both tests agree. No differencing needed for the level of returns. Models can be fitted directly to `quarter_return`.

Mean is white noise; variance is not

Diagnostic	p-value	Reject H_0?
Ljung–Box (lag 4)	0.93	no
Ljung–Box (lag 8)	0.998	no
Ljung–Box (lag 12)	0.99	no
Engle ARCH (lag 12)	**0.04**	**yes**

Implication. Whatever predictability exists lives in the *second moment*, not the first.

Train / test split and backtest design

- **Holdout split.** Train: 1981Q1 – 2022Q4 ($n = 168$). Test: 2023Q1 – 2026Q2 ($n = 14$).
- **Rolling-origin backtest.** Expanding window starting at 2018Q1, refit at every step, 34 one-step forecasts.
- All hyperparameters tuned on *training* data only, with `TimeSeriesSplit` (no random shuffling).
- Two horizons let us check the result is not an artifact of one test window.

Three naive benchmarks

- **Historical mean.** $\hat{y}_t = \bar{y}_{\text{train}}$. Optimal under i.i.d. returns.
- **Persistence (naive last).** $\hat{y}_t = y_{t-1}$. Optimal under a random walk.
- **Seasonal naive.** $\hat{y}_t = y_{t-4}$. Optimal if returns are perfectly periodic with period 4.

These set the bar that any “real” model must clear.

ARIMA and SARIMA grid search

ARIMA grid: $p \in \{0, \dots, 3\}$, $d \in \{0, 1\}$, $q \in \{0, \dots, 3\}$ (32 specifications).
SARIMA grid: same plus seasonal $(P, D, Q) \in \{0, 1\}^3$ at $s = 4$ (256 specifications).

Selected model	AIC	BIC
ARIMA(1, 0, 1)	5.90	18.40
SARIMA(0,1,2)$\times$(1,0,1)$_4$	4.01	19.38

Both choose low-order specifications — consistent with the flat ACF.

Lagged regression family

Design matrix (per quarter t):

$$x_t = (y_{t-1}, y_{t-2}, y_{t-3}, y_{t-4}, \mathbb{1}_{Q2}, \mathbb{1}_{Q3}, \mathbb{1}_{Q4})^\top$$

- **OLS** — baseline regression on the seven features.
- **Ridge** — ℓ_2 penalty, α grid 10^{-3} to 10^3 on a log scale.
- **LASSO** — ℓ_1 penalty, α grid 10^{-4} to 10^1.
- Hyperparameters tuned with `TimeSeriesSplit` (5 expanding folds), *not* random K-fold.

Evaluation framework

Point-error metrics:

- MAE — mean absolute error
- RMSE — root mean squared error
- sMAPE — symmetric MAPE (robust near zero)

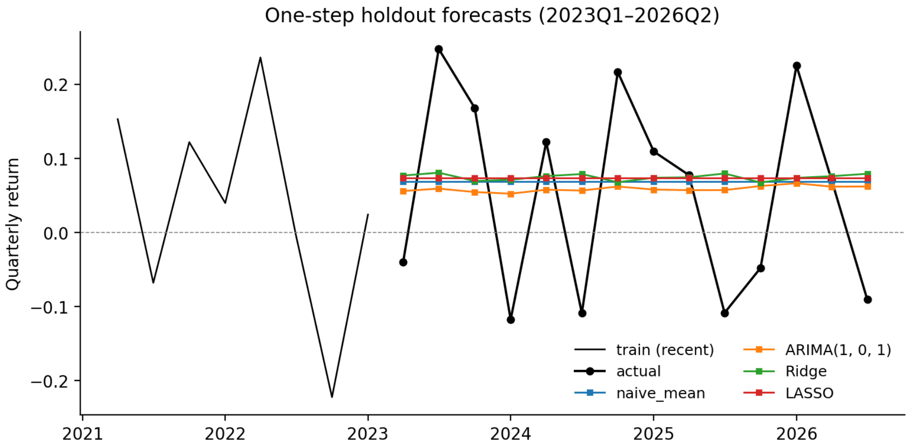
Significance test: Diebold–Mariano with the Harvey–Leybourne– Newbold (HLN) small-sample correction.

H_0: equal predictive accuracy versus the historical-mean benchmark. Two-sided p-value reported.

Holdout metrics: 14 quarters from 2023Q1

Model	MAE	RMSE	RMSE vs. mean
naive_mean (baseline)	0.115	0.131	0.0%
naive_last	0.178	0.201	−53.2%
naive_seasonal	0.165	0.208	−58.4%
ARIMA(1,0,1)	0.116	**0.129**	**+1.7%**
SARIMA(0,1,2)(1,0,1)$_4$	0.112	0.131	−0.0%
OLS	0.123	0.141	−7.1%
Ridge	0.116	0.133	−1.4%
LASSO	0.115	0.132	−0.5%

Forecasts on the holdout window



Black markers: realized quarterly returns. Coloured lines: one-step-ahead forecasts. Every model hugs the historical-mean line.

Diebold–Mariano test on the holdout

Model vs. `naive_mean`	DM stat	HLN p
ARIMA(1,0,1)	−0.79	0.46
SARIMA(0,1,2)(1,0,1)$_4$	0.00	0.998
OLS	1.20	0.27
Ridge	0.87	0.42
LASSO	0.56	0.60
naive_last	3.09	**0.011**
naive_seasonal	1.98	**0.078**

No "real" model significantly beats the historical mean. `naive_last` is significantly *worse*.

Robustness: rolling-origin backtest

- Expanding training window starting at 2018Q1.
- Refit every model at every quarter, predict one step ahead.
- 34 forecasts spanning pre-COVID, COVID-shock, and recent quarters.
- Goal: rule out the possibility that the holdout ranking is an artifact of a single test window.

Rolling-origin metrics: 34 quarters from 2018Q1

Model	MAE	RMSE	RMSE vs. mean
naive_mean (baseline)	0.126	0.155	0.0%
naive_last	0.198	0.237	−53.2%
naive_seasonal	0.179	0.228	−47.7%
ARIMA(1,0,1)	0.126	0.153	+0.7%
SARIMA(0,1,2)(1,0,1)$_4$	**0.122**	**0.153**	+**0.8**%
Ridge	0.127	0.156	−0.7%
LASSO	0.125	0.155	−0.0%

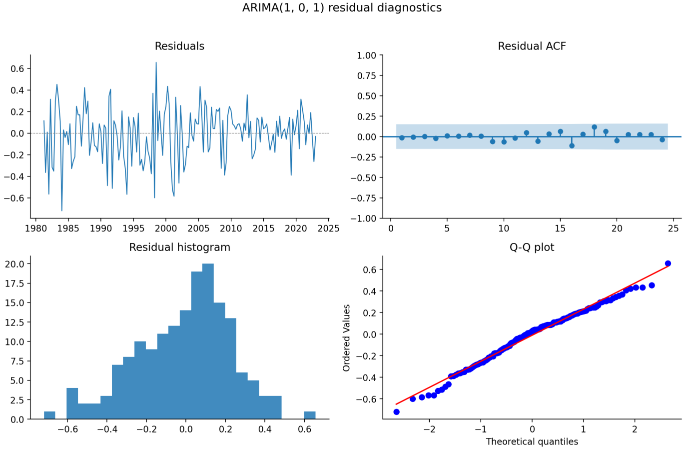
Same story: best model beats baseline by $\sim 1\%$, DM p around 0.7.

Why the regressions collapsed

Feature	OLS	Ridge ($\alpha = 10^3$)	LASSO ($\alpha = 0.083$)
lag 1	-0.007	-0.001	0.000
lag 2	-0.009	-0.002	0.000
lag 3	-0.004	-0.000	0.000
lag 4	-0.014	-0.001	0.000
Q2 dummy	-0.007	0.001	0.000
Q3 dummy	-0.035	-0.003	0.000
Q4 dummy	-0.024	-0.001	0.000

Ridge picks the largest α in the grid. LASSO drives every coefficient to exactly zero — it recovers the historical mean.

ARIMA residual diagnostics



Ljung–Box on residuals: $p > 0.99$ at all lags (white noise). **ARCH on residuals:** $p = 0.009$ **(still heteroskedastic).**

Answer to the research question

Verdict

No. Apple's quarterly returns over 1981–2026 do *not* contain temporal structure that translates into out-of-sample forecasts beating the historical mean.

Evidence that converges on the same answer:

- Flat ACF / PACF; Ljung–Box $p > 0.92$ at every tested lag.
- Best ARIMA improves RMSE by 1%–2%; DM $p \in [0.4, 0.7]$.
- Time-aware LASSO sets every coefficient to zero.
- Same conclusion on a single 14-quarter holdout and a 34-quarter rolling backtest.

Why this null result is informative

- Mean dynamics are statistically indistinguishable from white noise.
- Variance dynamics are demonstrably *not* white noise (ARCH $p < 0.05$ on raw and residuals).
- Predictability of Apple quarterly returns lives in the second moment, not the first.
- This is consistent with the efficient-market hypothesis on the mean and the volatility-clustering literature on the variance.

Limitations

- **Sample size.** 168 training quarters is a fundamental ceiling on statistical power.
- **Structural break.** The 2020Q1 COVID shock dominates a handful of rolling-origin forecasts.
- **Missing covariates.** Product-revenue shares are available only from 2009Q1, so they cannot be used over the full sample.
- **Single ticker.** Conclusions are about Apple, not about equities in general.

Future work & Q&A

Direct extensions:

- **GARCH(1,1)** on residuals — model the volatility structure we identified, evaluate value-at-risk forecasts.
- **State-space TVP models** that adapt to regime shifts (Markov-switching, time-varying parameters).
- **Earnings-window predictors** — analyst revisions and implied volatility ahead of each release.